Curriculum Vitae

Chirag Singhal
Department of Mathematics, Statistics, and Computer Science
University of Illinois at Chicago (UIC)
322 Science and Engineering Office
851 S Morgan St
Chicago, Illinois 60607-7045, USA
csingh23@uic.edu
https://sites.google.com/view/chiragsinghal/home

Personal Information

Indian Citizen, F-1 U.S. Visa

Education

University of Illinois, Chicago (UIC) *2022 – 2027 (Expected)*
Doctor of Philosophy (PhD) in Mathematics
- Thesis Advisor: Prof. Alina Carmen Cojocaru
- Thesis Topic: Arithmetic properties of Abelian Varieties and related objects

Indian Institute of Technology, Kanpur (IIT-K), India *2020 – 2022*
Master of Science (MSc) in Mathematics
- Thesis Advisor: Prof. Saurabh Kumar Singh
- Thesis topic: Modular Forms and Galois Representations

University of Warwick, UK *2017 – 2018*
Master of Science (MSc) in Mathematics
- Thesis Advisor: Prof. John Cremona
- Thesis topic: Point Counting Algorithms for Elliptic Curves over Finite Fields

International Institute of Information Technology Hyderabad, India *2013 – 2017*
Bachelor of Technology (BTech Hons.) in Computer Science and Engineering
- Project Advisor: Prof. Kannan Srinathan
- Project topic: Topics in Cryptography and Computational Number Theory

Research interests

Number Theory, Arithmetic Geometry, Computational Number Theory, Theory of Computation, Computational Complexity Theory and Cryptography.

In Particular: Galois Representations, Torsion of Elliptic Curves, Rational points on modular curves.

Research manuscripts and publications

2. C. Singhal, *Effective bounds on size of isogeny class of elliptic curves over number fields*, in preparation. *Feb 2026*

1. A.C. Cojocaru and C. Singhal, *Global liftings of infinite families of elliptic curves over finite fields*, in preparation. (25 pages) *Feb 2026*

Talks/Presentations

Conference Talks

3. *Computing Endomorphism Ring of Supersingular Elliptic curves* — *March 2026*
   Arizona Winter School '26
   University of Arizona, Tucson, AZ, USA

2. *Finiteness of Isolated j-invariants* — *May 2025*
   Arithmetic Geometry at UNT
   University of North Texas, Denton, TX, USA

1. *Global lifting of elliptic curves over finite fields* — *Nov 2024*
   Bayou Arithmetic Research Days (BARD) 5
   Xavier University of Louisiana, New Orleans, LA, USA

Seminar Talks

13. *Serre's Open Image Theorem* — *Jan 2026*
    UIC Graduate Number Theory Seminar
    University of Illinois Chicago, Chicago, IL, USA

12. *Modular Curves and Serre's Uniformity Conjecture* — *Sep 2025*
    UIC Graduate Number Theory Seminar
    University of Illinois Chicago, Chicago, IL, USA

11. *Overview of proof for Fermat's Last Theorem* — *April 2025*
    UIC Graduate Number Theory Seminar
    University of Illinois Chicago, Chicago, IL, USA

10. *Busy beaver proof paradigm and applications to Riemann Hypothesis* — *Mar 2025*
    UIC Math Club
    University of Illinois Chicago, Chicago, IL, USA

9. *Modular forms and Partition functions* — *Nov 2024*
   UIC Graduate Number Theory Seminar
   University of Illinois Chicago, Chicago, IL, USA

8. *Lifting elliptic curves over finite fields to elliptic curve over* $\mathbb{Q}$ — *Oct 2024*
   UIC Graduate Number Theory Seminar
   University of Illinois Chicago, Chicago, IL, USA

7. *Sieve methods* — *Sep 2024*
   UIC Graduate Number Theory Seminar
   University of Illinois Chicago, Chicago, IL, USA

6. *Intro to Drinfeld Modules* — *April 2024*
   UIC Graduate Number Theory Seminar
   University of Illinois Chicago, Chicago, IL, USA

5. *The field of definition of a Drinfeld module II* — *Mar 2023*
   UIC Number Theory Seminar
   University of Illinois Chicago, Chicago, IL, USA

4. *The field of definition of a Drinfeld module I* — *Mar 2023*
   UIC Number Theory Seminar
   University of Illinois Chicago, Chicago, IL, USA

3. *Bombieri-Stepanov proof of the Hasse-Weil bound* — *Nov 2022*
   UIC Number Theory seminar
   University of Illinois Chicago, Chicago, IL, USA

2. *Ramification in number fields* — *April 2022*
IIT-K Masters students' presentations seminar
Indian Institute of Technology-Kanpur, Kanpur, India

1. *Modular Forms and Galois Representations* — *Nov 2021*
IIT-K Masters students' presentations seminar
Indian Institute of Technology-Kanpur, Kanpur, India

Achievements and Awards

5. **Bernard Kurtin Scholarship:** $1000 scholarship by the MSCS (Mathematics, Statistics, and Computer Science) department at UIC. — *Dec 2024*

4. **Academic Performance awards at IIT-K**: Top mathematics student in 2021 as well as 2022. — *July 2022*
   - J.N. Kanpur Prize
   - Academic Excellence Award
   - General Proficiency Medal
   - Jasmine and Mohiuddin Merit Scholarship worth ₹10,000

3. **IIIT-H Dean's Merit List**: Top 4% of students academically. — *May 2017*

2. **Research Scholarships**:
   - S$5,400 at National University of Singapore, Singapore — *May 2017*
   - €3,000 at Lund University, Sweden — *May 2016*
   - ₹20,000 at Indian Statistical Institute Kolkata, India — *June 2015*

1. **Indian National Mathematics Olympiad (INMO)**: Qualified in INMO (Indian equivalent of USAMO) among 30 students and called for the IMO training camp for the final selection of India's IMO team. — *May 2011, May 2012*

Teaching

University of Illinois Chicago, USA

13. *MATH 210*: Teaching Assistant for *Calculus III* — *Spring 2026*
12. *MATH 294*: Course Instructor for *Special Topics in Mathematics* — *Fall 2025*
11. *MATH 310*: Grader for *Applied Linear Algebra* — *Fall 2025*
10. *MATH 310*: Teaching Assistant for *Applied Linear Algebra* — *Summer 2025*
9. *MATH 181*: Teaching Assistant for *Calculus II* — *Spring 2025*
8. *MATH 088*: Course Instructor for *Intermediate Algebra Workshop* — *Fall 2024*
7. *MATH 220*: Teaching Assistant for *Differential Equations* — *Summer 2024*
6. *MATH 160*: Teaching Assistant for *Finite Mathematics for Business* — *Spring 2024*
5. *MATH 514*: Guest Lecture for A*lgebraic Number Theory* — *Fall 2023*
4. *MATH 180*: Teaching Assistant for *Calculus I* — *Fall 2023*
3. *MATH 310*: Grader for *Applied Linear Algebra* — *Summer 2023*
2. *MATH 210*: Teaching Assistant for *Calculus III* — *Spring 2023*
1. *MATH 170*: Teaching Assistant for *Calculus for Life Sciences at UIC* — *Fall 222*

International Institute of Information Technology Hyderabad, India

3. *CSE 418*: Teaching Assistant for *Principles of Information Security* — *Spring 2017*
2. *IMA 404*: Teaching Assistant for *Number Theory and Cryptology* — *Fall 2016*
1. *CSE 311*: Teaching Assistant for *Formal Methods* — *Spring 2016*

Mentoring/Outreach

8. Mentored two middle school students, Ariana Romero and Christopher Sanchez, on a research project for QED, a school-level research math festival organised by Math Circles Chicago. The project is based on a variation of Bulgarian solitaire. — *Dec 2024*
7. Judge for a school-level research math festival QED, organised by Math Circles Chicago. — *Dec 2024*
6. Teacher for Math Circles Chicago, an organisation aimed at spreading mathematical interest among children around the age 12-17. — *August 2024* -ongoing.
5. Mentored Diana Dobrovolsky, an undergraduate student at University of Illinois Chicago(UIC), for a reading project on Quadratic Reciprocity through UIC's Directed Reading Program. — *Fall 2024*
4. Mentored Garam Choi, an undergraduate student at Colby College, for a reading project on analytic number theory (via the Twoples program). — *Fall 2022*
3. Co-organised and instructed for a week-long workshop on elementary Real Analysis for around 30 first-year undergrads. — *July 2021*
2. Mentored 6 students in the 3-week-long camp MTTS OFCM designed to inculcate mathematical thinking among first-year undergrads. — *Sep 2021*
1. Appointed as IIIT-H's Undergrad Math Help Cell Head. I was responsible to help students who needed extra help with mathematics. — *May 2016*

Service

Refereed for Journal of Number Theory — *Sep 2023*